

MTH166

Lecture-15

Method of Variation of Parameters

Topic:

Solution of Non-Homogeneous LDE with Constant coefficients.

Learning Outcomes:

To use Method of Variation of Parameters to solve Non-Homogeneous LDE with constant coefficients.

Method of Variation of Parameters:

Let us consider 2nd order Non-homogeneous LDE with constant coefficients as:

$$ay'' + by' + cy = r(x) \quad (1)$$

Let two solutions of C.F. $(aD^2 + bD + c) = 0$ be y_1 and y_2

$$\text{i.e. } y_c = c_1y_1 + c_2y_2$$

$$\text{Wronskian, } W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$\text{P.I. } y_p = -y_1 \int \frac{y_2 r(x)}{W} dx + y_2 \int \frac{y_1 r(x)}{W} dx$$

General solution is: $y = C.F. + P.I.$

$$\text{i.e. } y = y_c + y_p$$

Problem 1. Find the general solution of: $y'' + y = \sec x$

Solution: The given equation is:

$$y'' + y = \sec x \quad (1)$$

$$\text{S.F.: } (D^2 + 1)y = \sec x \quad \text{where } D \equiv \frac{d}{dx}$$

$$\Rightarrow f(D)y = r(x) \quad \text{where } f(D) = (D^2 + 1) \text{ and } r(x) = \sec x$$

To find Complimentary Function (C.F.):

$$\text{A.E.: } f(D) = 0 \quad \Rightarrow (D^2 + 1) = 0 \quad \Rightarrow D^2 = -1 \Rightarrow D = i, -i$$

$\therefore$ Complimentary function is given by:

$$y_c = e^{0x}(c_1 \cos x + c_2 \sin x)$$

$$\Rightarrow y_c = (c_1 \cos x + c_2 \sin x)$$

To find Particular Integral (P.I.):

Comparing y_c with $y_c = c_1 y_1 + c_2 y_2$

Here $y_1 = \cos x$, $y_2 = \sin x$

$$\text{Wronskian, } W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = (\cos^2 x + \sin^2 x) = 1$$

$$\text{P.I. is given by: } y_p = -y_1 \int \frac{y_2 r(x)}{W} dx + y_2 \int \frac{y_1 r(x)}{W} dx$$

$$\Rightarrow y_p = -\cos x \int \frac{\sin x (\sec x)}{1} dx + \sin x \int \frac{\cos x (\sec x)}{1} dx$$

$$\Rightarrow y_p = -\cos x \int \tan x dx + \sin x \int dx \quad \left(\sec x = \frac{1}{\cos x} \right)$$

$$\Rightarrow y_p = -\cos x (-\log|\cos x|) + \sin x (x)$$

$\therefore$ General solution is given by: $y = \text{C.F.} + \text{P.I.}$

$$\text{i.e. } y = y_c + y_p$$

$$\Rightarrow y = (c_1 \cos x + c_2 \sin x) + \cos x (\log|\cos x|) + x \sin x \quad \textbf{Answer.}$$

Polling Question:

The Wronskian of $y'' + 5y' + 4y = 18e^{2x}$ is:

(A) $3e^{5x}$

(B) $-3e^{-5x}$

(C) $3e^{-5x}$

(D) *None of these.*

Problem 2. Find the general solution of: $y'' + y = \operatorname{cosec} x$

Solution: The given equation is:

$$y'' + y = \operatorname{cosec} x \quad (1)$$

$$\text{S.F.: } (D^2 + 1)y = \operatorname{cosec} x \quad \text{where } D \equiv \frac{d}{dx}$$

$$\Rightarrow f(D)y = r(x) \quad \text{where } f(D) = (D^2 + 1) \text{ and } r(x) = \operatorname{cosec} x$$

To find Complimentary Function (C.F.):

$$\text{A.E.: } f(D) = 0 \quad \Rightarrow (D^2 + 1) = 0 \quad \Rightarrow D^2 = -1 \Rightarrow D = i, -i$$

$\therefore$ Complimentary function is given by:

$$y_c = e^{0x}(c_1 \cos x + c_2 \sin x)$$

$$\Rightarrow y_c = (c_1 \cos x + c_2 \sin x)$$

To find Particular Integral (P.I.):

Comparing y_c with $y_c = c_1 y_1 + c_2 y_2$

Here $y_1 = \cos x$, $y_2 = \sin x$

$$\text{Wronskian, } W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = (\cos^2 x + \sin^2 x) = 1$$

$$\text{P.I. is given by: } y_p = -y_1 \int \frac{y_2 r(x)}{W} dx + y_2 \int \frac{y_1 r(x)}{W} dx$$

$$\Rightarrow y_p = -\cos x \int \frac{\sin x (\operatorname{cosec} x)}{1} dx + \sin x \int \frac{\cos x (\operatorname{cosec} x)}{1} dx$$

$$\Rightarrow y_p = -\cos x \int dx + \sin x \int \cot x dx \quad \left(\operatorname{cosec} x = \frac{1}{\sin x} \right)$$

$$\Rightarrow y_p = -\cos x(x) + \sin x (\log|\sin x|)$$

$\therefore$ General solution is given by: $y = \text{C.F.} + \text{P.I.}$

$$\text{i.e. } y = y_c + y_p$$

$$\Rightarrow y = (c_1 \cos x + c_2 \sin x) + \sin x (\log|\sin x|) - x \cos x \quad \textbf{Answer.}$$

Problem 3. Find the general solution of: $y'' + y = \tan x$

Solution: The given equation is:

$$y'' + y = \tan x \quad (1)$$

$$\text{S.F.: } (D^2 + 1)y = \tan x \quad \text{where } D \equiv \frac{d}{dx}$$

$$\Rightarrow f(D)y = r(x) \quad \text{where } f(D) = (D^2 + 1) \text{ and } r(x) = \tan x$$

To find Complimentary Function (C.F.):

$$\text{A.E.: } f(D) = 0 \Rightarrow (D^2 + 1) = 0 \Rightarrow D^2 = -1 \Rightarrow D = i, -i$$

$\therefore$ Complimentary function is given by:

$$y_c = e^{0x}(c_1 \cos x + c_2 \sin x)$$

$$\Rightarrow y_c = (c_1 \cos x + c_2 \sin x)$$

To find Particular Integral (P.I.):

Comparing y_c with $y_c = c_1 y_1 + c_2 y_2$

Here $y_1 = \cos x$, $y_2 = \sin x$

$$\text{Wronskian, } W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = (\cos^2 x + \sin^2 x) = 1$$

$$\text{P.I. is given by: } y_p = -y_1 \int \frac{y_2 r(x)}{W} dx + y_2 \int \frac{y_1 r(x)}{W} dx$$

$$\Rightarrow y_p = -\cos x \int \frac{\sin x (\tan x)}{1} dx + \sin x \int \frac{\cos x (\tan x)}{1} dx$$

$$\Rightarrow y_p = -\cos x \int \frac{\sin^2 x}{\cos x} dx + \sin x \int \sin x dx \quad \left(\tan x = \frac{\sin x}{\cos x} \right)$$

$$\Rightarrow y_p = -\cos x \int \frac{1 - \cos^2 x}{\cos x} dx + \sin x (-\cos x)$$

$$\Rightarrow y_p = -\cos x \int (\sec x - \cos x) dx - \sin x \cos x$$

$$\Rightarrow y_p = -\cos x \log |\sec x + \tan x| + \cos x \sin x - \sin x \cos x$$

$\therefore$ General solution is given by: $y = \text{C.F.} + \text{P.I.}$

$$\text{i.e. } y = y_c + y_p$$

$$\Rightarrow y = (c_1 \cos x + c_2 \sin x) - \cos x (\log |\sec x + \tan x|) \quad \textbf{Answer.}$$

Problem 4. Find the general solution of: $y'' + y = \cot x$

Solution: The given equation is:

$$y'' + y = \cot x \quad (1)$$

$$\text{S.F.: } (D^2 + 1)y = \cot x \quad \text{where } D \equiv \frac{d}{dx}$$

$$\Rightarrow f(D)y = r(x) \quad \text{where } f(D) = (D^2 + 1) \text{ and } r(x) = \cot x$$

To find Complimentary Function (C.F.):

$$\text{A.E.: } f(D) = 0 \quad \Rightarrow (D^2 + 1) = 0 \quad \Rightarrow D^2 = -1 \Rightarrow D = i, -i$$

$\therefore$ Complimentary function is given by:

$$y_c = e^{0x}(c_1 \cos x + c_2 \sin x)$$

$$\Rightarrow y_c = (c_1 \cos x + c_2 \sin x)$$

To find Particular Integral (P.I.):

Comparing y_c with $y_c = c_1 y_1 + c_2 y_2$

Here $y_1 = \cos x$, $y_2 = \sin x$

$$\text{Wronskian, } W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = (\cos^2 x + \sin^2 x) = 1$$

$$\text{P.I. is given by: } y_p = -y_1 \int \frac{y_2 r(x)}{W} dx + y_2 \int \frac{y_1 r(x)}{W} dx$$

$$\Rightarrow y_p = -\cos x \int \frac{\sin x (\cot x)}{1} dx + \sin x \int \frac{\cos x (\cot x)}{1} dx$$

$$\Rightarrow y_p = -\cos x \int \cos x dx + \sin x \int \frac{\cos^2 x}{\sin x} dx \quad \left(\cot x = \frac{\cos x}{\sin x} \right)$$

$$\Rightarrow y_p = -\cos x (\sin x) + \sin x \int \frac{1 - \sin^2 x}{\sin x} dx$$

$$\Rightarrow y_p = -\cos x (\sin x) + \sin x \int (\operatorname{cosec} x - \sin x) dx$$

$$\Rightarrow y_p = -\cos x (\sin x) + \sin x (-\log |\operatorname{cosec} x + \cot x|) + \sin x (\cos x)$$

$\therefore$ General solution is given by: $y = \text{C.F.} + \text{P.I.}$

$$\text{i.e. } y = y_c + y_p$$

$$\Rightarrow y = (c_1 \cos x + c_2 \sin x) - \sin x (\log |\operatorname{cosec} x + \cot x|) \quad \textbf{Answer.}$$

Polling Question:

The Wronskian of $y'' + 9y = \cos 3x$ is:

(A) 1

(B) 3

(C) -3

Problem 5. Find the general solution of: $y'' - 2y' - 3y = e^x$

Solution: The given equation is:

$$y'' - 2y' - 3y = e^x \quad (1)$$

S.F.: $(D^2 - 2D - 3)y = e^x$ where $D \equiv \frac{d}{dx}$

$\Rightarrow f(D)y = r(x)$ where $f(D) = (D^2 - 2D - 3)$ and $r(x) = e^x$

To find Complimentary Function (C.F.):

A.E.: $f(D) = 0 \Rightarrow (D^2 - 2D - 3) = 0 \Rightarrow (D - 3)(D + 1) = 0$

$\Rightarrow D = 3, -1$ (real and distinct roots)

$\therefore$ Complimentary function is given by:

$$y_c = c_1 e^{3x} + c_2 e^{-1x}$$

To find Particular Integral (P.I.):

Comparing y_c with $y_c = c_1 y_1 + c_2 y_2$

Here $y_1 = e^{3x}$, $y_2 = e^{-x}$

$$\text{Wronskian, } W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{3x} & e^{-x} \\ 3e^{3x} & -e^{-x} \end{vmatrix} = (-e^{3x}e^{-x} - 3e^{3x}e^{-x}) = -4e^{2x}$$

$$\text{P.I. is given by: } y_p = -y_1 \int \frac{y_2 r(x)}{W} dx + y_2 \int \frac{y_1 r(x)}{W} dx$$

$$\Rightarrow y_p = -e^{3x} \int \frac{e^{-x}(e^x)}{-4e^{2x}} dx + e^{-x} \int \frac{e^{3x}(e^x)}{-4e^{2x}} dx$$

$$\Rightarrow y_p = \frac{e^{3x}}{4} \int e^{-2x} dx - \frac{e^{-x}}{4} \int e^{2x} dx$$

$$\Rightarrow y_p = \frac{e^{3x}}{4} \left(\frac{e^{-2x}}{-2} \right) - \frac{e^{-x}}{4} \left(\frac{e^{2x}}{2} \right) = -\frac{1}{4} e^x$$

$\therefore$ General solution is given by: $y = \text{C.F.} + \text{P.I.}$

$$\text{i.e. } y = y_c + y_p$$

$$\Rightarrow y = (c_1 e^{3x} + c_2 e^{-1x}) - \frac{1}{4} e^x \quad \textbf{Answer.}$$

